COMMON SQL DATA TYPES

Concepts • Categories • SQL Commands • Practical Examples

Learning Outcomes

By the end of this lesson, students should be able to:

- Explain why data types are required when creating database tables
- Classify common SQL data types into major categories
- Choose suitable data types for numbers, text, dates, Boolean and binary data
- Write CREATE TABLE and INSERT statements using appropriate data types
- Identify common data-type selection mistakes

What Is a SQL Data Type?

A data type defines what kind of value a column can store.

Examples:

- Age → whole number
- Employee name → text
- Joining date → date
- Salary → exact decimal value

Category	Common types
Numeric	INT, BIGINT, DECIMAL, FLOAT
Character / String	CHAR, VARCHAR, TEXT
Date and Time	DATE, TIME, DATETIME, TIMESTAMP
Boolean	BOOLEAN / BOOL
Binary	BINARY, VARBINARY, BLOB

Note: Exact type names and supported features vary among MySQL, Oracle, SQL Server and PostgreSQL.

1. Numeric Data Types

Use numeric types for quantities, counts, measurements and money.

Data type	Use	Example
TINYINT	Very small whole number	1
SMALLINT	Small whole number	250
INT / INTEGER	General whole number	25000
BIGINT	Very large whole number	9876543210
DECIMAL(p,s)	Exact decimal value	12500.50
FLOAT / DOUBLE	Approximate decimal value	23.75

Important choice

- Use INT for age or quantity.
- Use DECIMAL for salary and financial amounts.
- Use FLOAT/DOUBLE for approximate scientific measurements.

```
salary DECIMAL(10,2)
```

2. Character / String Data Types

Use string types to store names, codes, addresses and text.

Data type	Description	Example
CHAR(n)	Fixed-length text	'BD'
VARCHAR(n)	Variable-length text	'Rahim'
TINYTEXT	Short text	Short note
TEXT	Text content	Address
MEDIUMTEXT	Larger text	Article
LONGTEXT	Very large text	Long document

CHAR vs VARCHAR

CHAR(2) is suitable for fixed-length codes.

VARCHAR(100) is suitable for names because the stored text length can vary.

```
    employee_name VARCHAR(100),
branch_code CHAR(3)
```

3. Date and Time Data Types

Use these types to store dates, times and timestamps.

Data type	Stored value example
DATE	'2026-10-31'
TIME	'11:30:00'
DATETIME	'2026-10-31 11:30:00'
TIMESTAMP	'2026-10-31 11:30:00'
YEAR	2026

Example columns

```
          joining_date DATE,
start_time TIME,
created_at DATETIME
```

Use DATE for birth or joining dates. Use DATETIME when both date and time are needed.

4. Boolean and 5. Binary Data Types

Boolean represents a true/false condition; binary types store bytes.

Type	Purpose	Example
BOOLEAN / BOOL	True or false flag	TRUE
BINARY(n)	Fixed-length bytes	Binary code
VARBINARY(n)	Variable-length bytes	Binary value
BLOB	Binary Large Object	Image / file

MySQL note

In MySQL, BOOLEAN is a synonym for TINYINT(1). Commonly, 0 represents FALSE and 1 represents TRUE.

```
            is_active BOOLEAN,
profile_photo BLOB
```

SQL Command: CREATE TABLE

Define each column with an appropriate data type.

```
CREATE TABLE Employee (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100) NOT NULL,
    age INT,
    salary DECIMAL(10,2),
    joining_date DATE,
    is_active BOOLEAN
);
```

What does it do?

- Creates a table named Employee
- Defines six columns
- Assigns a suitable type to each column
- Sets employee_id as the primary key

SQL Commands: INSERT and SELECT

Insert sample values, then retrieve the stored records.

```
INSERT INTO Employee
(employee_id, employee_name, age,
 salary, joining_date, is_active)
VALUES
(101, 'Rahim', 28, 45000.50,
 '2026-01-15', TRUE);
```

```
SELECT *
FROM Employee;
```

Expected result (sample)

ID	Name	Salary	Active
101	Rahim	45000.50	1

More SQL Commands: ALTER and UPDATE

Modify a table definition or update stored values.

Add a new column

```
ALTER TABLE Employee
ADD email VARCHAR(120);
```

Update a value

```
UPDATE Employee
SET salary = 50000.00
WHERE employee_id = 101;
```

Best practice: Always use a WHERE condition in UPDATE and DELETE statements when you intend to change only selected rows.

How to Choose the Right Data Type

Select the type based on the real-world meaning and expected range of the value.

Data to store	Recommended type	Reason
Employee ID	INT	Whole-number identifier
Employee name	VARCHAR(100)	Variable-length text
Salary	DECIMAL(10,2)	Exact financial value
Date of birth	DATE	Date only
Account active?	BOOLEAN	True / false condition
Photo / document	BLOB	Binary file content

Avoid storing numbers as text when you need arithmetic, sorting or numeric comparisons.

MySQL and Oracle: Common Type Equivalents

The concepts are similar, but some type names differ.

Purpose	MySQL	Oracle
Whole / decimal numbers	INT, DECIMAL(p,s)	NUMBER(p,s)
Variable-length text	VARCHAR(n)	VARCHAR2(n)
Fixed-length text	CHAR(n)	CHAR(n)
Date and time	DATE, DATETIME, TIMESTAMP	DATE, TIMESTAMP
Large text	TEXT / LONGTEXT	CLOB
Binary large object	BLOB	BLOB

Oracle note: NUMBER is commonly used for numeric columns; VARCHAR2 is preferred over VARCHAR in Oracle.

Classroom Practice

Create a Student table using suitable data types.

Task: Choose data types for the following columns.

Column	Requirement
student_id	Unique whole-number ID
student_name	Maximum 100 characters
cgpa	Exact value with 2 decimal places
admission_date	Date of admission
is_enrolled	True / false

Suggested solution

```
CREATE TABLE Student (
  student_id INT PRIMARY KEY,
  student_name VARCHAR(100),
  cgpa DECIMAL(3,2),
  admission_date DATE,
  is_enrolled BOOLEAN
);
```

Extension: Insert two student records and retrieve them using SELECT * FROM Student;

Quick Review

- Numeric → INT, BIGINT, DECIMAL, FLOAT
- String → CHAR, VARCHAR, TEXT
- Date & Time → DATE, TIME, DATETIME, TIMESTAMP
- Boolean → BOOLEAN / BOOL
- Binary → BINARY, VARBINARY, BLOB
- CREATE TABLE defines columns; INSERT adds rows; SELECT retrieves rows